

Educations

Ph.D. in Electrical and Computer Engineering GPA: 4.0/4.0	Sept. 2022 - Present
Georgia Institute of Technology (Gatech), Atlanta, GA, USA	
B.S. in Electrical Engineering GPA: 4.06/4.30	Sept. 2017 - July 2022
National Taiwan University (NTU), Taipei, Taiwan	

Research Experience

Graduate Research Assistant	
SCCAD (former GTCAD), Georgia Institute of Technology, Advisor: Prof. Sung Kyu Lim	Sept. 2022 - Present
3D IC Optimization	Fall 2024 - Present
o Timing optimization through reduced critical-path snaking and inter-die cell relocation (Snake-3D).	
o Developed an RC-calibrated differentiable clock-tree optimization framework for 3D ICs, incorporating interconnect RC effects into timing-aware clock-tree optimization (CLK-3D).	
Semiconductor Research Corporation (SRC) – JUMP 2.0 CHIMES	Fall 2023 - Spring 2024
o Bridged the PPA gap between legacy-node 3D ICs and advanced-node 2D ICs.	
o Reduced WNS of legacy-node 3D ICs by 50–93% compared to SOTA, achieving 87.9–97.7% of advanced-node 2D performance.	
Die Bonding Bumps and Pads Legalization for 3D ICs	Sept. 2022 - May 2023
o Developed a 3D-via legalization stage with a refinement step while minimizing runtime impact; see the related TCAD journal article and ISPD conference paper .	
o Eliminated all 3D-via overlaps when via utilization is less than 40%.	
Individual Study	
EDA Lab., National Taiwan University, Advisor: Prof. Yao-Wen Chang	Apr. 2020 - Sept. 2021
Independent Study on Physical Design	
o Participated in physical-design-related global contests three times across different teams and placed in the top five twice (ISPD contest 2021 and ICCAD contest 2021).	
Individual Study	
ALCom Lab., National Taiwan University, Advisor: Prof. Jie-Hong Jiang	Sept. 2020 - June 2021
Independent Study on Reversible Circuit Synthesis	
o Implemented a reversible circuit synthesizer for controlled-NOT (CNOT) gates.	

Work Experience

R&D Engineer Intern	Sunnyvale, CA
Synopsys Inc.	May 2025 - Dec. 2025
o Investigated machine-learning-based signal-integrity (SI) analysis for 2.5D ICs, evaluating multiple models to predict final-stage SI performance from early-stage design data.	
o Independently implemented two provided prototypes in C++ and merged the resulting code into the main branch.	
Software Engineer R&D Intern	Hsinchu, Taiwan
Synopsys Inc. - Concurrent Clock-and-Data (CCD) Team	July 2021 - Aug. 2021
o Enhanced the flip-flop merging algorithm in Synopsys IC Compiler II (ICC II) by considering the flip-flop’s timing (slack)	
o Improved the timing quality in 13 real industrial benchmarks: 5.14%/3.47% in worst/total negative slack	
Software Engineer R&D Intern	Hsinchu, Taiwan
Cadence Inc. - Conformal Team	June 2020 - Sept. 2020
o Provided electronic change order (ECO) test cases in a 5-person team for Cadence Conformal’s robustness test	
o Generated 2000+, 5 kinds of cases in which Conformal acts unusually and proposed 3 improved flows to fix	

Publication

- o **Yen-Hsiang Huang** and Sung Kyu Lim, "CLK-3D: RC-Calibrated Differentiable Clock-Tree Optimization for 3D ICs," IEEE/ACM International Conference on Computer-Aided Design, 2026.
- o **Yen-Hsiang Huang** and Sung Kyu Lim, "Snake-3D: Differentiable Learning for Cross-Tier Logic Path Snaking Optimization in 3D ICs," IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2025. [PDF]
- o **Yen-Hsiang Huang**, Sai Pentapati, Anthony Agnesina, Moritz Brunion, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024. [PDF]
- o Sai Pentapati, Anthony Agnesina, Moritz Brunion, **Yen-Hsiang Huang**, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs", ACM International Symposium on Physical Design (ISPD), 2023. [PDF]

Honors and Awards

Honorable Mention Award in the ACM ISPD contest 2021	March 2021
o 4th place worldwide in the contest.	
TSMC scholarship for undergraduate student	Sept. 2020 - Jan. 2021
o Awarded to approximately 20 undergraduate students in Taiwan every semester.	
The Dean’s List Award	Spring 2020
o Ranked 1st, with 4.30/4.30 GPA in the semester.	